

USER ENGAGEMENT ANALYSIS FOR RESTAURANT SUCCESS

About Yelp:

Yelp is a web and mobile platform that functions as a crowd-sourced local business review site. Users can submit reviews, photos, and tips about businesses while also browsing information and ratings left by others.

Business Problem:

In a competitive market like the restaurant industry, understanding the factors that influence business success is crucial for stakeholders. Utilizing the Yelp dataset, this project aims to investigate the relationship between user engagement (reviews, tips, and check-ins) and business success metrics (review count, ratings) for restaurants.

Data Analysis:

Analysis and Findings:

average_review_count	104.097789
min_review_count	5.000000
max_review_count	7568.000000
median_review_count	15.000000
average_star_rating	3.523969
min_star_rating	1.000000
max_star_rating	5.000000
median_star_rating	3.500000

- Out of 150K businesses, 35K are restaurant businesses and are open.
- The above table shows the distribution of business success metrics (Review count and Average ratings)

Highest Review counts:

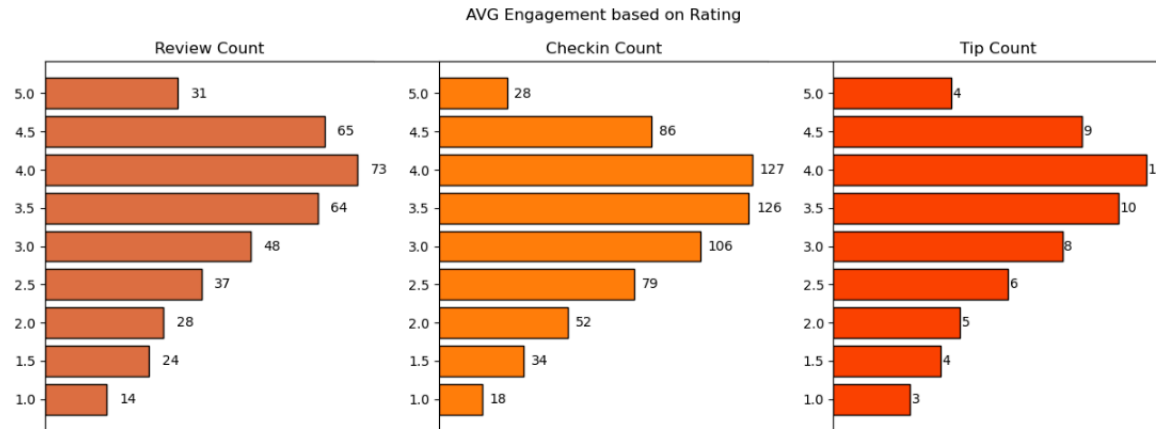
	name	review_count	avg_rating
0	McDonald's	16490	1.868702
1	Chipotle Mexican Grill	9071	2.381757
2	Taco Bell	8017	2.141813
3	Chick-fil-A	7687	3.377419
4	First Watch	6761	3.875000
5	Panera Bread	6613	2.661905
6	Buffalo Wild Wings	6483	2.344828
7	Domino's Pizza	6091	2.290210
8	Wendy's	5930	2.030159
9	Chili's	5744	2.514706

Highest Ratings:

	name	review_count	avg_rating
0	ā café	48	5.0
1	two birds cafe	77	5.0
2	the brewers cabinet production	13	5.0
3	taqueria la cañada	17	5.0
4	la bamba	44	5.0
5	la 5th av tacos	24	5.0
6	el sabor mexican and chinese food	21	5.0
7	eat.drink.Om...YOGA CAFE	7	5.0
8	d4 Tabletop Gaming Cafe	8	5.0
9	cabbage vegetarian cafe	12	5.0

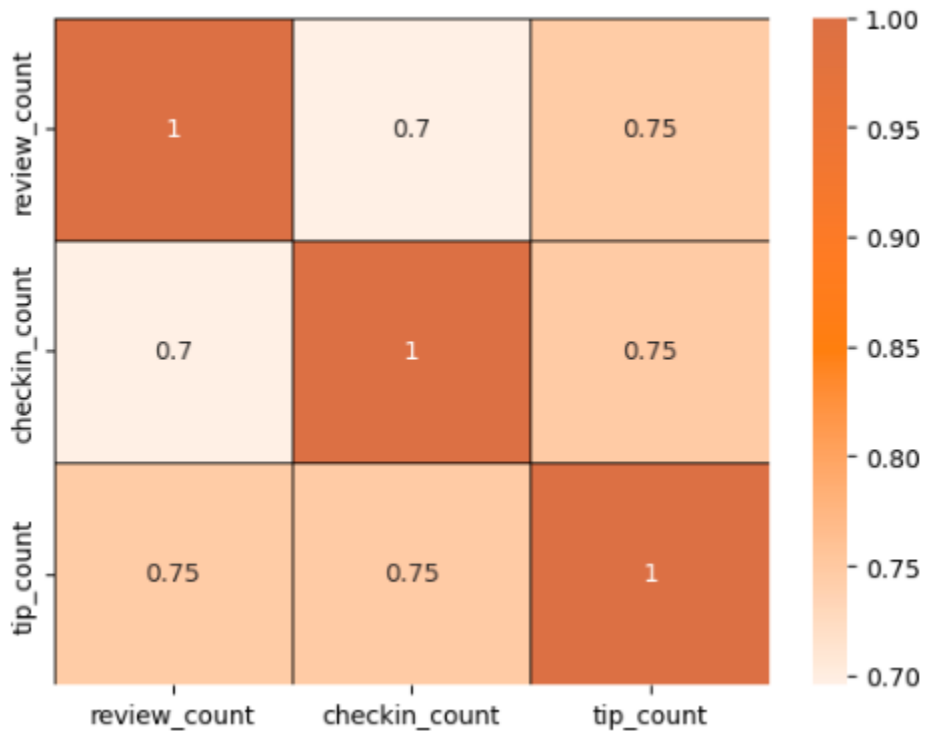
- Highest ratings do not guarantee a higher review count or vice versa.
- Success of restaurants is not solely determined by ratings or review counts.
- Review count reflects user engagement but not necessarily overall customer satisfaction or business performance.

Do restaurants with higher engagement tend to have higher ratings?



- Data shows a general increase in average review, check-in, and tip counts as ratings improve from 1 to 4 stars.
- Restaurants rated 4 Stars exhibit the higher engagement and show a downward trend for rating above 4.
- The drop in engagement at 5 stars might suggest either a saturation point where fewer customers feel compelled to add their reviews or a selectivity where only a small, satisfied audience frequents these establishments.

Correlation between the number of reviews, tips, and check-ins for a business?



- These correlations suggest that user engagement across different platforms (like reviews, tips, and check-ins) is interlinked. Higher activity in one area tends to be associated with higher activity in others.
- Businesses should focus on strategies that boost all types of user engagement, as increases in one type of engagement are likely to drive increases in others, enhancing overall visibility and interaction with customers.

Is there a difference in the user engagement (reviews, tips, and check-ins) between high-rated and low-rated businesses?

	review_count	checkin_count	tip_count
rating_category			
High-Rated	63.099378	80.71859	8.069794
Low-Rated	37.152862	64.84321	5.456341

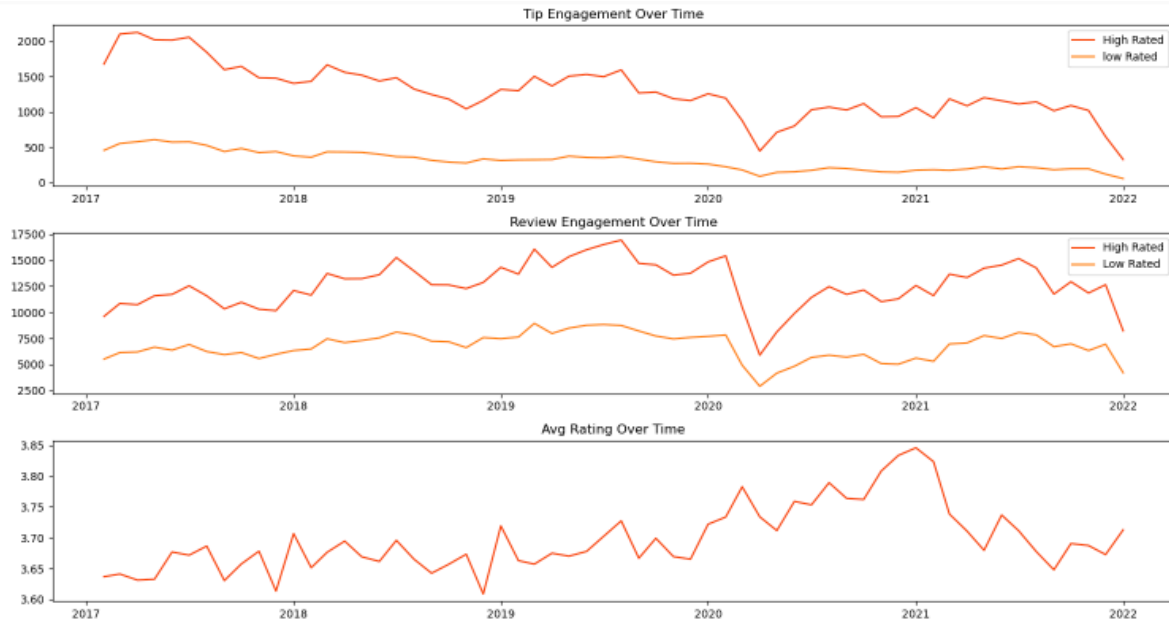
- It clearly indicates a clear correlation between higher ratings and increased user engagement across reviews, tips and check-Ins.
- This pattern underscores the importance of maintaining high service and quality standards as these appear to drive more reviews, check-ins and tips which are critical metrics of customer engagement and satisfaction.

How do the success metrics (review_count or avg_rating) of restaurants vary across different states and cities?

Success Score = avg_rating * log(review_count + 1)

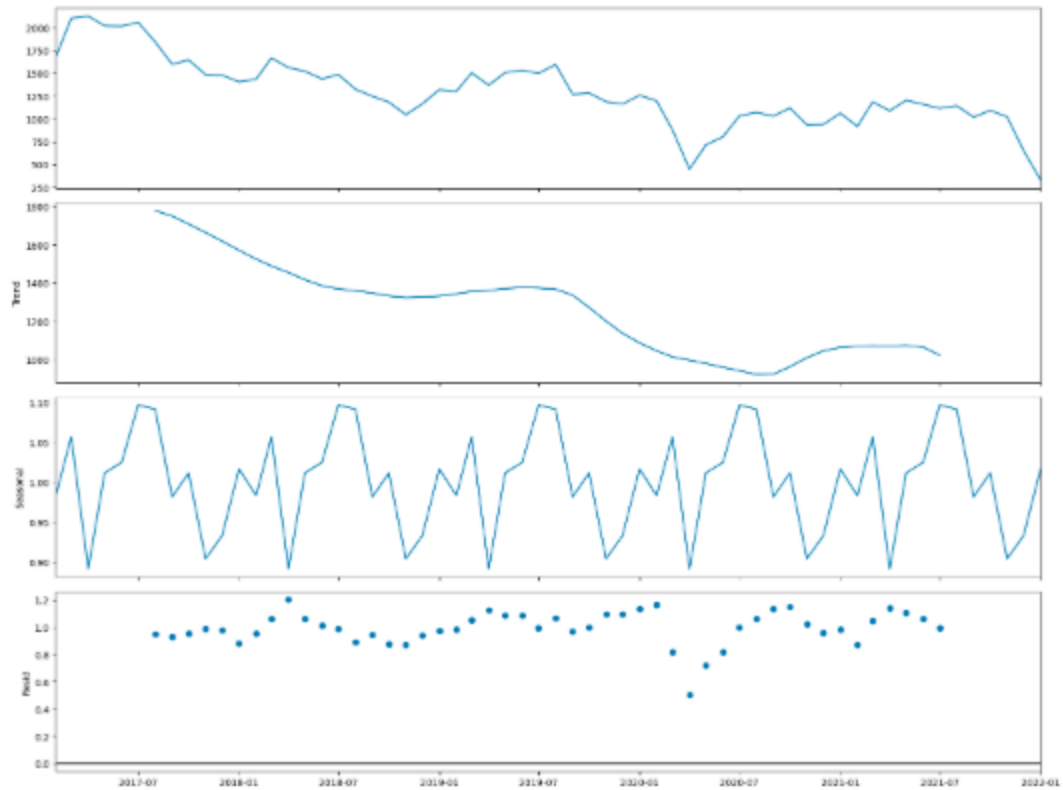
- Philadelphia emerges as the top city with the highest success score, indicating a combination of high ratings and active user engagement.
- Following Philadelphia, Tampa, Indianapolis and Tucson rank among the top cities with significant success scores, suggesting thriving restaurant scenes in these areas.

Are there any patterns in user engagement over time for successful businesses compared to less successful ones?

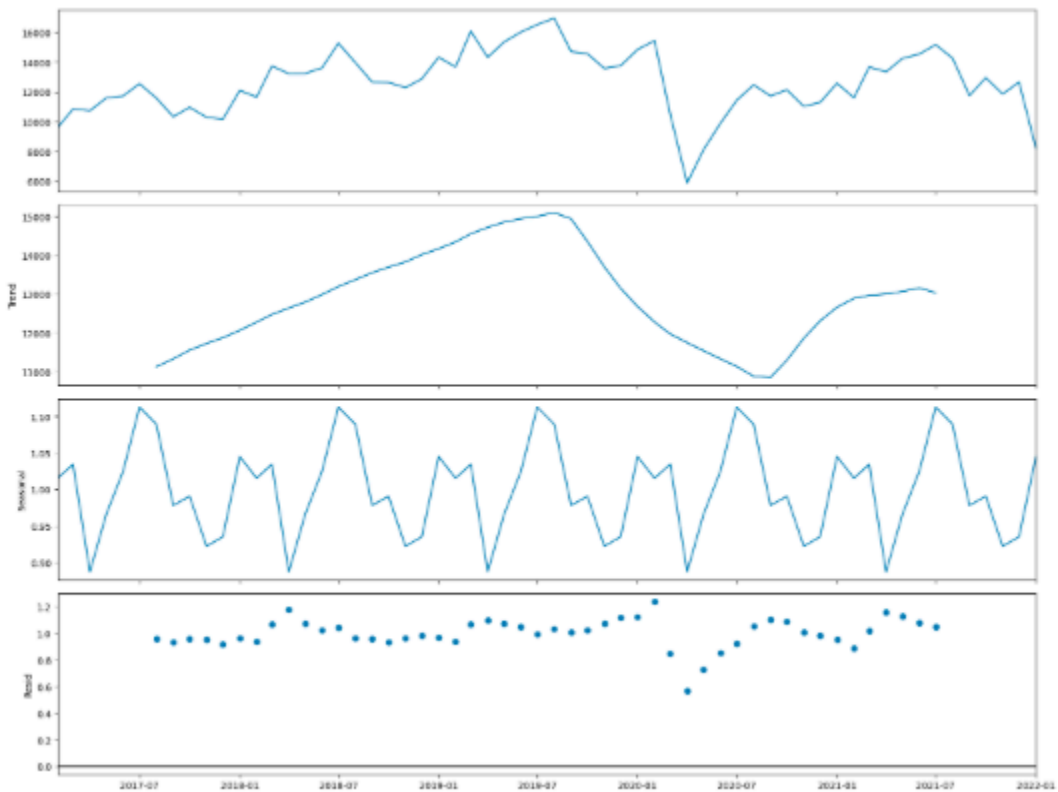


- Successful businesses, particularly those with higher ratings(> 3.5), exhibit consistent and possibly increasing use engagement over the time.
- High rated restaurants maintain a steady or growing level of user engagement over the time, reflecting ongoing customer interest and satisfaction.

Tip count Trend: with downward trend

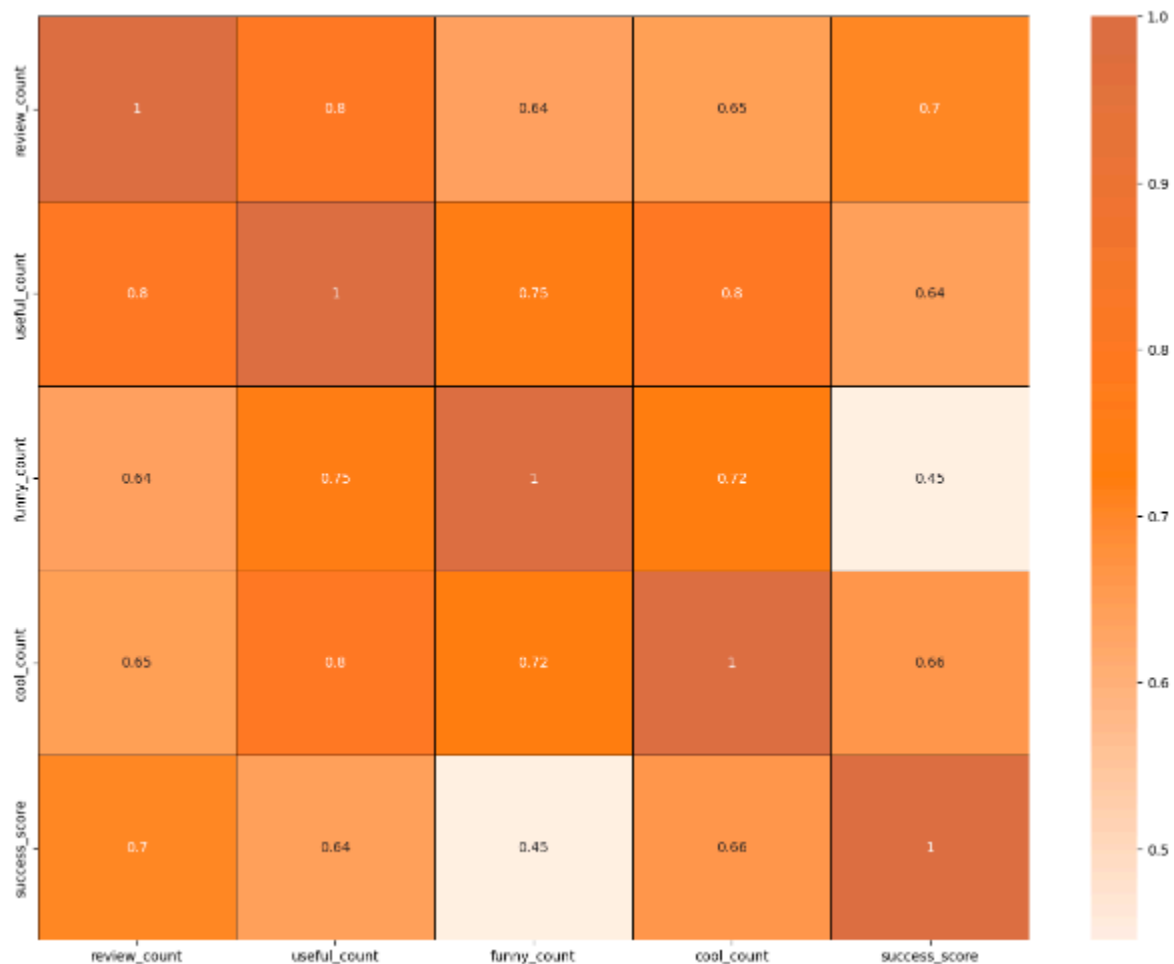


Review count trend: - with upward trend



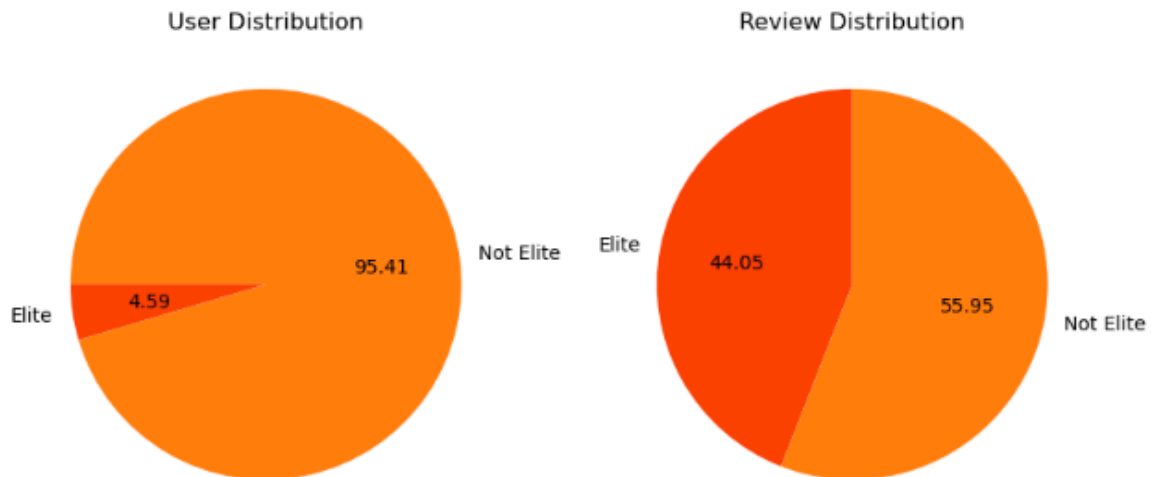
- Starting and Ending years (NOV - MAR) is highly engaging.

How does the sentiment of reviews and tips (useful, funny, cool) correlate with the success metrics of restaurants?



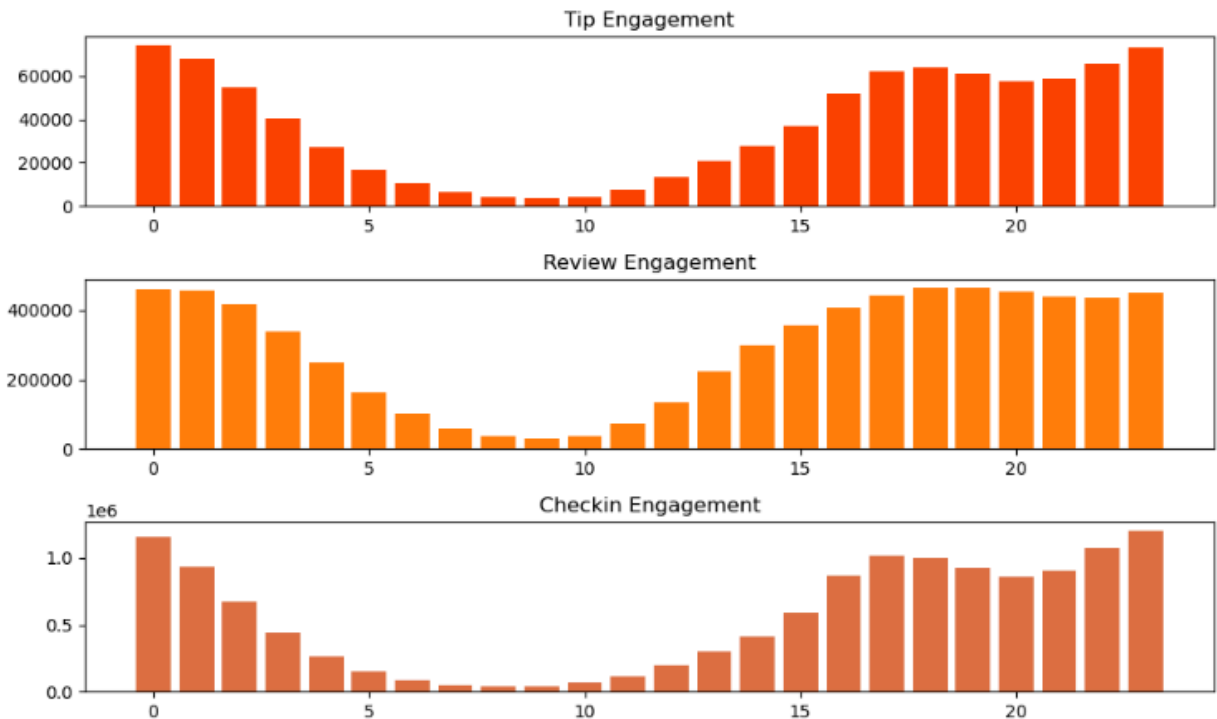
- **"Useful" "funny, " and "cool"** are attributes associated with user reviews. They represent the feedback provided by users about the usefulness, humor, or coolness of a particular review.
- **Higher counts of useful, funny, and cool reviews suggest greater user engagement and satisfaction**, which are key factors contributing to a restaurant's success.

Is there any difference in engagement of elite users and non elite users?



- Elite users are individuals who have been recognized and awarded the "Elite" status by Yelp for their active and high-quality contributions to the platform, such as frequent and detailed reviews, photos, and check-ins, among other criteria.
- Elite users, despite being significantly fewer in number, contribute a substantial proportion of the total review count compared to non-elite users.
- Elite users often provide detailed and insightful reviews, which can influence other users' perceptions and decisions regarding a business.
- Reviews from elite users may receive more attention and visibility on the Yelp platform due to their status, potentially leading to higher exposure for businesses.
- Establishing a positive relationship with elite users can lead to repeat visits and loyalty, as they are more likely to continue supporting businesses they have had good experiences with

Busiest Hours:



- The busiest hours for restaurants, based on user engagement, span from 4 pm to 1 am.
- Knowing the peak hours allows businesses to optimize their staffing levels and resource allocation during these times to ensure efficient operations and quality service delivery.
- The concentration of user engagement during the evening and night hours suggests a higher demand for dining out during these times, potentially driven by factors such as work schedules, social gatherings, and leisure activities.

Recommendations

- Utilizing insights from the analysis of various metrics such as user engagement, sentiment of reviews, peak hours, and the impact of elite users, businesses can make informed decisions to drive success.
- Understanding customer preferences, behavior, and satisfaction levels is paramount. Businesses should focus on delivering exceptional experiences to meet customer expectations.
- By leveraging data on peak hours and user engagement, businesses can optimize staffing levels, resource allocation, and operating hours to ensure efficiency and quality service delivery during high-demand periods.
- Positive reviews from elite users and high user engagement can boost a business's online visibility and reputation. Maintaining active engagement with customers and responding promptly to feedback is crucial for building credibility and attracting new customers.
- Collaborating with elite users and leveraging their influence can amplify promotional efforts, increase brand awareness, and drive customer acquisition. Building strong relationships with

key stakeholders, including loyal customers, can further strengthen a business's position in the market.

- Businesses can adjust their operating hours or introduce special promotions to capitalize on the increased demand during peak hours.
- Less successful businesses may need to focus on strategies to enhance user engagement over time, such as improving service quality, responding to customer feedback.
- Cities with high success scores present opportunities for restaurant chains to expand or invest further.